

Rory Schepp

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Education

MSc, Mathematics, Dalhousie University 2025–present (expected April 2027)

- Thesis: generators and relations for $O(\mathbb{Q}, n)$, the group of rational orthogonal matrices. Supervisor: Dr. Neil J. Ross.

BSc (Honours), Mathematics, Dalhousie University 2021–2025

- First-class honours; GPA 4.02 / 4.30; Dean's List every term.
- Honours thesis: *Methods to Improve Lattice Surgery Compilation*. Supervisors: Dr. Christopher Dean and Dr. Peter Selinger.

Research interests

Generators and relations for groups; presentations of orthogonal matrix groups; quantum circuit synthesis and compilation; ZX-calculus; lattice surgery; reversible circuits.

Research experience

Master's thesis research, Dalhousie University. Supervisor: Dr. Ross 2025–present

- Studying $O(\mathbb{Q}, n)$, the group of $n \times n$ orthogonal matrices with rational entries.
- Building on its generation by matrices that each change at most four coordinates, working out the relations among these generators.

Summer research assistant, Dalhousie University Summer 2025

Supervisor: Dr. Christopher Dean; funded by Dr. Peter Selinger.

- With the honours thesis, developed a ZX-calculus framework for compiling quantum circuits to lattice surgery.
- Defined the ZX-calculus DAG and lattice surgery compatibility; showed every Clifford+ T circuit compiles to a compatible diagram.
- Determined which standard ZX rewrite rules preserve compatibility, giving compilers a safe set of simplifications.
- Outlined a procedure placing operations in a 3D space-time lattice using approximate Steiner trees.

Summer research assistant, Dalhousie University Summer 2024

Co-supervisors: Dr. Dongho Lee and Dr. Christopher Dean; funded by Dr. Selinger.

- Independent project on generators and relations for reversible Boolean and ternary circuits.
- Investigated whether finitely many relations suffice for known finite generating sets, using computer search to find structure.

Presentations

* denotes presenting author.

1. **Schepp, R.***, Dean, C. (Nov. 2025). *Methods for Improving Compilation of Quantum Circuits to Lattice Surgery*. Dalhousie Mathematics and Statistics Graduate Seminar, Halifax, NS.

2. Dean, C.*, **Schepp, R.** (Oct. 2025). *ZX-Spider Packing*. AMS Special Session on New Horizons for Quantum Error Correction: Categorical Approaches to Topological Codes, AMS Fall Eastern Virtual Sectional Meeting (online).
3. **Schepp, R.***, Dean, C. (Oct. 2025). *Methods for Improving Compilation of Quantum Circuits to Lattice Surgery*. Science Atlantic Mathematics, Statistics and Computer Science Conference. Third place, graduate oral presentations.
4. **Schepp, R.*** (Oct. 2024). *Equivalence Relations on Reversible Circuits*. Science Atlantic Mathematics, Statistics and Computer Science Conference.

Awards and funding

2025	Science Atlantic Graduate Research Award, Third Place, Oral Presentation (Mathematics, Statistics and Computer Science Conference)
2022–2025	Dalhousie Entrance Renewable Scholarship
2021–2025	Dean's List, every term

Teaching experience

Teaching assistant and marker, Department of Mathematics and Statistics, Dalhousie University Sept. 2025–present

Term	Course	Role
Fall 2025, Fall 2026	MATH 1000, Calculus	Prepare problems, grade quizzes
Fall 2025, Fall 2026	Mathematics Learning Centre	Drop-in shifts across six first- and second-year courses
Winter 2026	MATH 1030, Linear Algebra	Quiz grading; invigilated two midterms and two finals
Winter 2026	MATH 4116, Cryptography	Marker, weekly assignments
Fall 2025	MATH 3070, Number Theory	Marker, weekly assignments

Service

Vice-President Internal, Dalhousie Mathematics and Statistics Graduate Society Oct. 2025–Oct. 2026